

Adaptive Echo Cancellation of Audio Signals Using Adaptive Filters

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Abstract—Echo cancellation is one of the critical tasks that needs to be accomplished when there is speech and audio transmission as the reflected signals create noise. This paper will present the comparative analysis of adaptive filters LMS, NLMS, and RLS for the purpose of echo cancellation. The reason behind the development of this paper is to conduct an analysis of the three mentioned approaches in terms of performance in the case of echo cancellation. For this purpose, each approach will be considered under different parameters and filter orders and then analyzed in terms of their efficiency in terms of output and errors created.

Index Terms—Echo cancellation, adaptive filters, LMS, NLMS, RLS, speech and audio communication, parameter analysis, filter order, error reduction

I. INTRODUCTION

Echo cancellation plays a vital role in speech and audio communication systems since echo signals interfere with the signal of interest and decrease the signal quality. In most cases, the microphone input contains both the desired speech and echo. This causes distortion and decreases the effectiveness of the system output. Adaptive filters have been used to solve this issue since they are capable of adapting the weights according to the input signal [1].

In this work, three adaptive filters have been compared: LMS, NLMS, and RLS. The comparison has been conducted by varying the filter order and parameters in order to evaluate their performance. The objective of this work is to investigate which approach performs better in reducing echo with less error and quicker convergence [3].

II. OBJECTIVE

The major goal of this project is to analyze and compare three types of adaptive filters, LMS, NLMS, and RLS, in order to determine their efficiency in echo cancellation tasks. In particular, this analysis will evaluate the ability of each filter

type to extract an echo-free signal from the noisy input signal. In addition, this paper will investigate the impact of varying the order of the filter and different values of its parameters on the performance of each filter.

III. METHODOLOGY

A. Adaptive Filters Used

There were three adaptive filters applied in this research: LMS, NLMS, and RLS. All the three methods are widely used in echo cancellation applications due to the capability of adapting the filter coefficients depending on the input and reference signals. LMS is easy to realize, NLMS is more stable than LMS in case when the normalization of the input power is performed, while RLS offers faster adaptation and higher tracking accuracy. The same input signal was considered to make the comparison results unbiased [1], [3].

The main error signal can be described as:

$$e(n) = d(n) - y(n) \quad (1)$$

where $d(n)$ is the desired signal and $y(n)$ is the filter output. For LMS, the weight update equation is:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e(n)\mathbf{x}(n) \quad (2)$$

For NLMS, the update equation is:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu}{\|\mathbf{x}(n)\|^2 + \epsilon} e(n)\mathbf{x}(n) \quad (3)$$

For RLS, the update form is:

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n)e(n) \quad (4)$$

where $\mathbf{k}(n)$ is the gain vector. These equations describe how each algorithm adjusts its coefficients to reduce the echo error over time.

B. Filter Order Variation

First, there is the filter order that was manipulated in order to understand what effect it has on the performance of the algorithms. In this paper, some filter orders have been tested, including $M = 32, 64, 128, 256$, etc. Small orders cannot model the echo path efficiently, but large ones can. But too large orders might affect the speed of computations [3].

Each algorithm output was analyzed under different values of M . In order to see which algorithm performed better, the error values have been estimated.

For this purpose, Mean Square Error has been used:

$$\text{MSE} = \frac{1}{N} \sum_{n=1}^N e^2(n) \quad (5)$$

The Root Mean Square Error was also used:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N e^2(n)} \quad (6)$$

and the Mean Absolute Error was calculated as:

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |e(n)| \quad (7)$$

These formulas help measure how close the output is to the desired signal. Lower values mean better echo cancellation [1].

C. Step-Size / Parameter Variation

Once the appropriate order of the filter has been selected, the value of step-size or tuning parameters has been altered to analyze its impact on algorithm performance. The μ value was altered for LMS, the normalized step-size for NLMS, and the values of λ and δ for RLS were changed. It is necessary since selection of parameter affects not only the rate of convergence but also the performance of the algorithm [2], [3].

Generally, the small step-size leads to a stable operation, but slow convergence; at the same time, the large value allows achieving faster convergence at the expense of an increase in the errors or loss of stability. In RLS algorithms, the forgetting factor close to one leads to better memory of input signals. The purpose of this step has been to find optimal parameter values allowing reducing echo without distorting the speech signal [1].

Again, the identical measures have been chosen for calculating error. It has allowed assessing the quality of the algorithms based not only on output but also on error criteria.

IV. EXPERIMENTAL SETUP

The adaptive echo canceller was simulated by feeding the same speech signal to all the algorithms. After feeding the speech signal, the echo path was used to generate its reflection. Then LMS, NLMS, and RLS algorithms were used to suppress the echo. A block diagram of the system is shown in Fig. 1.

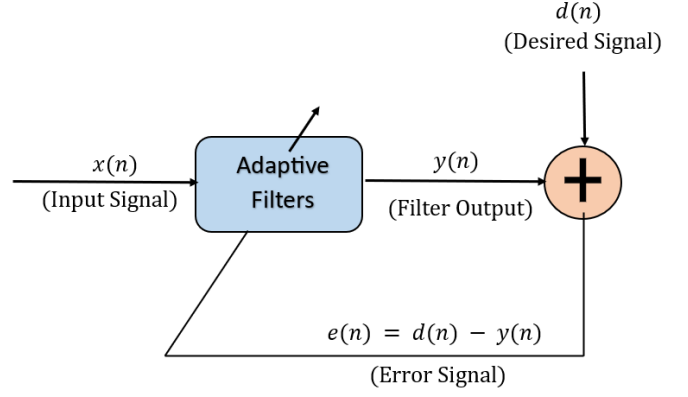


Fig. 1. Block diagram of adaptive echo cancellation system.

A. Input Signal

Same speech input signal was used in all three algorithms to ensure an equal comparison. This signal had both the useful speech and undesirable reflections, and it was ideal for echo cancellation experiments. The objective was to determine how effectively each algorithm could extract the clean speech signal from this mixture. Equal inputs in all cases ensured that the difference between the outcomes was purely due to the algorithm.

B. Echo Signal

An echo channel has been employed to replicate the behavior of delayed reflection in real-world communications systems. As can be seen in such scenarios, speech waves tend to get delayed by reflecting off various surfaces, causing the delayed version of the original wave to get propagated. For this reason, multiple copies of the original wave are generated, with delays and scaling applied, to form the echoed signal [8].

These waves are shown in Fig. 2. As can be seen in this figure, the echoed wave contains multiple replicas of the original wave, delayed at different time intervals. Therefore, it can be seen how the wave gets distorted and spreads.

C. Parameter Values

The first step involved varying the filter order to determine its impact on the performance. Filter orders used included $M = 32, 64, 128$, and 256 . Once a good filter order had been determined, the tuning parameters were individually varied for each adaptive filter algorithm, namely, LMS, NLMS, and RLS. In LMS algorithm, the value of the step size μ was varied. In NLMS algorithm, the value of the normalized step size and ϵ was varied [2].

D. Performance Measures

Performance of each algorithm was examined through Train MSE, Test MSE, Full MSE, RMSE, and MAE. The lower values of these errors suggest that the output is closer to the expected output signal without echo. This implies better cancellation and recovery of speech.

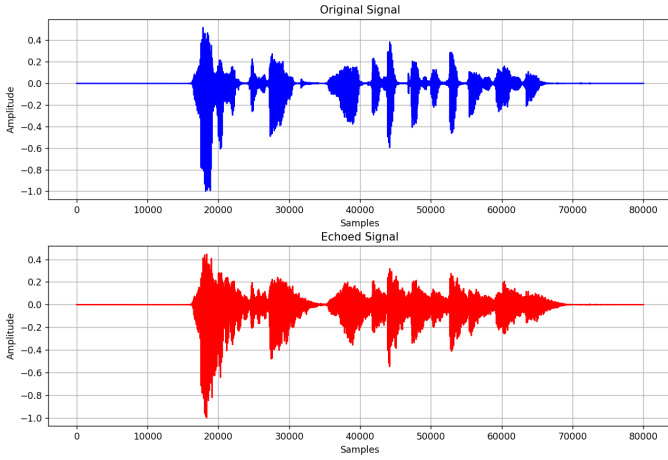


Fig. 2. Original and echoed speech signals.

Moreover, visual analysis of the output signal revealed the level of echo in the output signal.

V. SOFTWARE IMPLEMENTATION

The suggested model will be developed using Python programming language with the help of numeric and signal processing packages.

Firstly, the incoming speech signal will be captured using a sampling frequency of 16 kHz for a period of 5 seconds. The input signal will then be processed to remove the DC offset and normalize the amplitude of the incoming speech signal to stabilize it during processing.

To simulate realistic acoustic conditions, several replicas of the incoming speech signal will be generated with varying delays and attenuation levels. This resulting echoed speech signal will be fed as input $x(n)$, whereas the original incoming speech signal will be considered as the desired signal $d(n)$ [1].

A total of 70% of the entire dataset will be allocated for adaptive filter training purposes, whereas the remaining 30% will be reserved for testing the performance of the developed model.

Three adaptive filter designs, namely Least Mean Square (LMS), Normalized Least Mean Square (NLMS), and Recursive Least Square (RLS), are realized. These filters update their parameters through iterations in the training process and calculate the filtered and error responses.

- The LMS filter updates its weights using a constant step size.
- The NLMS filter incorporates a normalizing factor for enhanced performance.
- The RLS filter applies a recursive updating scheme with a forgetting parameter.

Measures like Mean Square Error (MSE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) are calculated for comparative purposes.

Lastly, the output is normalized, exported to audio files, and played for subjective testing. Plots showing the input

signal, echoed signal, and LMS, NLMS, and RLS responses are produced.

VI. RESULTS AND DISCUSSION

A. Comparison for Different M Values

The findings indicate that the order of the filter has a significant influence on the adaptive filters' performance. With smaller values of M , there was poor modeling of the echo path, hence resulting in large amounts of output errors [3]. With the increase in the order of the filter, the performance increased, and thus the amount of residual echoes decreased. The best results were achieved with $M=256$, hence its selection.

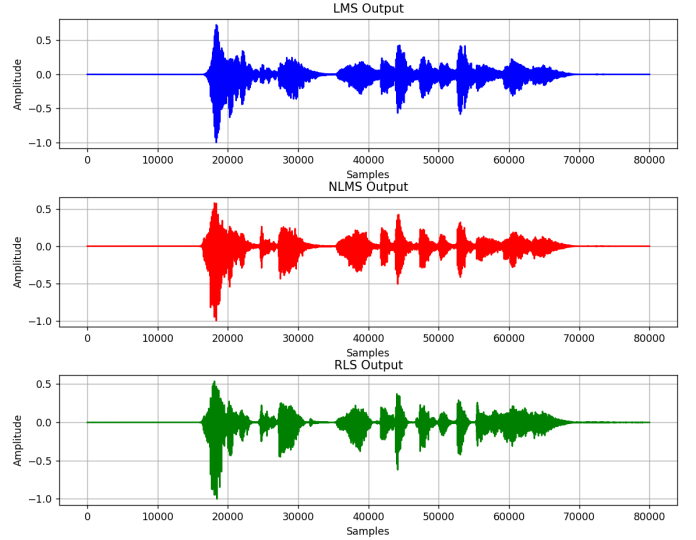


Fig. 3. Output comparison of LMS, NLMS, and RLS for $M = 256$.

From Fig. 3, it is observed that $M = 256$ yields the best performance for all three algorithms used. Though there is much distortion in the output of the LMS algorithm, the output of the NLMS algorithm has an improvement, but a little echo is remaining, while the output of the RLS algorithm has no distortions and has less reflected echoes [5].

From the increase in filter order, the adaptive filters can model the echo path effectively. From the diagram above, the output performance when using $M = 256$ becomes stable compared to other lower values of filter order. From the three algorithms, the output of the RLS performs well [1].

B. Effect of Filter Length (M) on MSE Performance

The performance of the algorithms with respect to filter length is shown in Fig. 1.

The LMS algorithm has a high MSE in comparison with other algorithms in any value of M ; hence, it has slow convergence and low performance. Although the MSE of LMS reduces as M changes from 32 to 256, the decrease in MSE is not very significant [5].

NLMS outperforms LMS due to its consistent decrease in MSE. The normalization process of NLMS results in reliable convergence with low computational complexity.

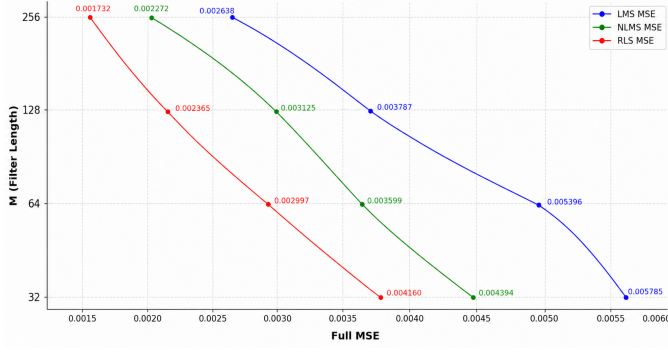


Fig. 4. Effect of filter length (M) on full MSE for LMS, NLMS, and RLS algorithms.

RLS has the smallest MSE among all algorithms and is capable of fast convergence even at small values of M. However, the improvement in performance causes a great computational load.

In general, MSE reduces as M becomes larger for all three algorithms. Based on the above analysis, RLS outperforms other algorithms in terms of accuracy; however, NLMS follows RLS, and LMS is the worst.

C. Comparison for Different Parameter Values

It is seen from the above graph that the output varies when the values of the algorithmic parameters are changed. In LMS, the output has more oscillation and noise, whereas in NLMS, it has less oscillation, and in RLS, the output is least oscillated.

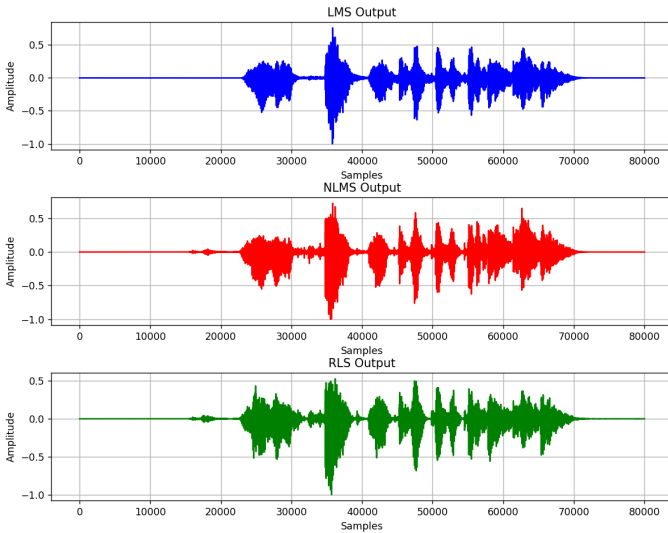


Fig. 5. Output comparison of LMS, NLMS, and RLS for different parameter values.

1) *LMS Parameter Analysis*: Fig. 6 illustrates the impact of the step size μ on the effectiveness of the LMS algorithm. As indicated in the graph, the MSE reduces initially before reaching a minimum at $\mu=0.01$. Beyond this point, the MSE starts rising because of instabilities caused by large values of μ [8].

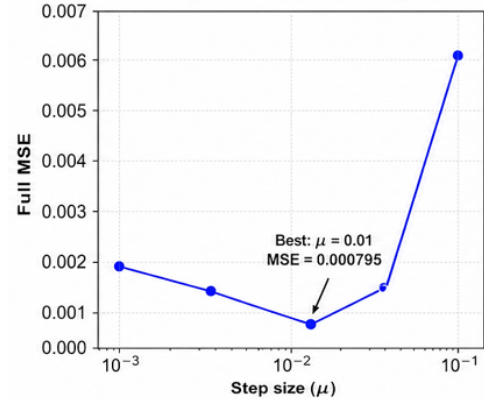


Fig. 6. Effect of step size (μ) on LMS performance.

2) *NLMS Parameter Analysis*: Fig. 7 illustrates the impact of step size μ on the NLMS algorithm. The algorithm is quite stable and gives low MSE for all cases except for $\mu=0.8$, which gives the optimum results. The NLMS is more stable and converges faster than LMS.

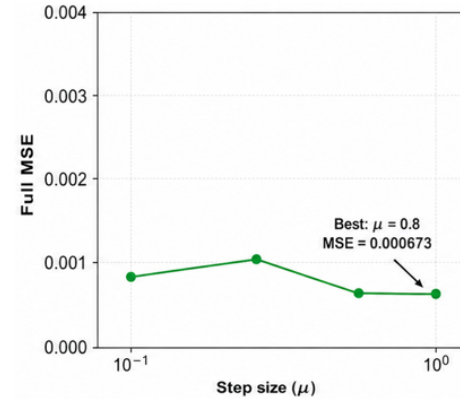


Fig. 7. Effect of step size (μ) on NLMS performance.

3) *RLS Parameter Analysis*: The performance of the RLS algorithm in relation to the forgetting factor λ is shown in Fig. 8. The optimum MSE is given by $\lambda=0.98$. A slight variation from this value gives a drastic fall in MSE [5].

As a conclusion, NLMS is stable and performs well. On the other hand, RLS gives the optimum performance and is highly sensitive to parameters. LMS is unstable and takes more time to converge.

The best values of parameters for which the overall MSE is minimized for each algorithm are provided in Table I.

TABLE I
PARAMETER EFFECT ON FULL MSE

Algorithm	Parameter	Best Value	Full MSE (Best)
LMS	Step size (μ)	0.01	0.000795
NLMS	Step size (μ)	0.8	0.000673
RLS	Forgetting factor (λ)	0.98	0.001154

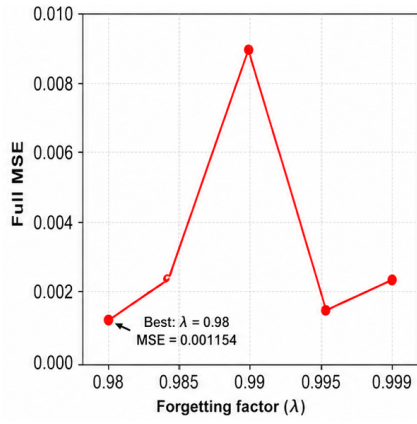


Fig. 8. Effect of forgetting factor (λ) on RLS performance.

D. Best Performing Algorithm

Echo cancellation is one of the crucial jobs to be performed in speech and audio communication channels, as the reflected sound tends to cause interference. This paper provides a comparative analysis of three types of adaptive filters used for echo cancellation, namely LMS, NLMS, and RLS algorithms.

The main objective of this paper is to analyze the behavior of these three techniques in echo cancellation under various parameters and orders of filters. Each technique was tested for its effectiveness and performance in reducing the echo from the input signal [1].

Performance of each algorithm was analyzed on the basis of output and errors generated. As observed, there are significant differences among the techniques based on their performance.

E. Output Signal Observation

It is observed from the output signals that there is a distinct difference between the three adaptive filters. For LMS, it can be seen that the speech signal is recovered but the amount of echo and wave distortion can still be found [2]. The use of NLMS helps in minimizing unwanted fluctuations and thus, produces an output with a clearer speech signal compared to LMS. It is noted that RLS provides the best quality output signal with little echo.

From the waveforms obtained above, it can be deduced that the adaptive filters were able to reduce the amount of echo but to different levels of effectiveness. In this case, it is evident that the output produced by RLS more or less replicates the desired output by suppressing echoes [3].

VII. CONCLUSION

Echo cancellation is one of the crucial jobs to be performed in speech and audio communication channels, as the reflected sound tends to cause interference. This paper provides a comparative analysis of three types of adaptive filters used for echo cancellation, namely LMS, NLMS, and RLS algorithms. The main objective of this paper is to analyze the behavior of these three techniques in echo cancellation under various parameters and orders of filters. Each technique was tested for

its effectiveness and performance in reducing the echo from the input signal. Performance of each algorithm was analyzed on the basis of output and errors generated. Among the three algorithms, RLS provides best performance with the lowest MSE and fastest convergence, followed by NLMS, which offers a good balance between accuracy and stability. LMS shows comparatively higher error and slower convergence, making it the least efficient among the three. Future work may include testing the adaptive echo cancellation system on real-time speech signals in a live environment. The performance can also be improved by implementing hybrid or optimized adaptive algorithms. Additionally, advanced techniques such as deep learning-based echo cancellation may be investigated for further enhancement.

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